LangChain overview

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LangChain is the easy way to start building completely custom agents and applications powered by LLMs. With under 10 lines of code, you can connect to OpenAI, Anthropic, Google, and more. LangChain provides a prebuilt agent architecture and model integrations to help you get started quickly and seamlessly incorporate LLMs into your agents and applications.

LangChain vs. LangGraph vs. Deep Agents

If you are looking to build an agent, we recommend you start with Deep Agents which comes "batteries-included", with modern features like automatic compression of long conversations, a virtual filesystem, and subagent-spawning for managing and isolating context.

Use LangGraph, our low-level agent orchestration framework and runtime, when you have more advanced needs that require a combination of deterministic and agentic workflows and heavy customization.

Get started

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Philosophy

Core components

Agents

Models

Messages

Tools

Short-term memory

Streaming

Structured output

LangChain agents are built on top of LangGraph in order to provide durable execution, streaming, human-in-the-loop, persistence, and more. You do not need to know LangGraph for basic LangChain agent usage.

We recommend you use LangChain if you want to quickly build agents and autonomous applications.

Create an agent

```
# pip install -qU langchain "langchain[anthropic]"
from langchain.agents import create_agent

def get_weather(city: str) -> str:
    """Get weather for a given city."""
    return f"It's always sunny in {city}!"

agent = create_agent(
    model="claude-sonnet-4-6",
    tools=[get_weather],
    system_prompt="You are a helpful assistant",
)

# Run the agent
agent.invoke(
    {"messages": [{"role": "user", "content": "what is the weather in
)
```

See the Installation instructions and Quickstart guide to get started building your own agents and applications with LangChain.

Use LangSmith to trace requests, debug agent behavior, and evaluate outputs. Set `LANGSMITH_TRACING=true` and your API key to get started.

Core benefits

Standard model interface

Different providers have unique APIs for interacting with models, including the format of responses. LangChain standardizes how you interact with models so that you can seamlessly swap providers and avoid lock-in.

Learn more ›

Easy to use, highly flexible agent

LangChain's agent abstraction is designed to be easy to get started with, letting you build a simple agent in under 10 lines of code. But it also provides enough flexibility to allow you to do all the context engineering your heart desires.

Learn more ›

Built on top of LangGraph

LangChain's agents are built on top of LangGraph. This allows us to take advantage of LangGraph's durable execution, human-in-the-loop support, persistence, and more.

Learn more ›

Debug with LangSmith

Gain deep visibility into complex agent behavior with visualization tools that trace execution paths, capture state transitions, and provide detailed runtime metrics.

Learn more ›

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Connect these docs to Claude, VSCode, and more via MCP for real-time answers.

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